

Sticky Hidden Markov Modeling of Comparative Genomic Hybridization

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Abstract—We develop a sticky hidden Markov model (HMM) with a Dirichlet distribution (DD) prior, motivated by the problem of analyzing comparative genomic hybridization (CGH) data. As formulated the sticky DD-HMM prior is employed to infer the number of states in an HMM, while also imposing state persistence. The form of the proposed hierarchical model allows efficient variational Bayesian (VB) inference, of interest for large-scale CGH problems. We compare alternative formulations of the sticky HMM, while also examining the relative efficacy of VB and Markov chain Monte Carlo (MCMC) inference. To validate the formulation, example results are presented for an illustrative synthesized data set and our main application—CGH, for which we consider data for breast cancer. For the latter, we also make comparisons and partially validate the CGH analysis through factor analysis of associated (but distinct) gene-expression data.

Index Terms—, DNA copy number, hidden Markov model (HMM), hierarchical Bayesian modeling, multi-task learning (MTL), variational Bayes (VB).

I. INTRODUCTION

COMPARATIVE genomic hybridization (CGH) yields data consisting of fluorescence intensity ratios of test and reference DNA samples. The intensity ratios provide information about the number of DNA copies in localized regions of a chromosome. Array CGH data analysis has recently attracted increasing interest in both the biology and statistics communities. Specifically, there is a growing need for algorithms that can automatically identify gains and losses in number of copies, and relate this to disease and illness.

A number of well-known methods strive to fulfill this need. For example, [1], [20], [21], [23], and [24] use segmentation to identify chromosomal segments with altered copy number. A variation of a binary segmentation method [23], called circular binary segmentation (CBS), segments the CGH data in each chromosome and computes the within-segment means. An edge filter is applied in [21] to detect segments. Since there is also a clear dependence among the intensity ratios of neighboring clones, [20] performs smoothing using the signs of neighboring

data values, inspecting the width and magnitude of the segments to detect regions of copy number change. A disadvantage of such methods is that they cannot directly detect gains or losses. A recent paper by [35] applies penalized matrix decomposition (PMD) for selecting important “clones” in array CGH data. Nevertheless, the sequential information is not explicitly exploited in such a matrix decomposition method, and the biological meaning of the selected “clones” are not assigned via this method.

To make use of the physical dependence of the nearby fragments or “clones,” the hidden Markov model (HMM) has been utilized by [12], [15] to analyze array CGH data, of which [12] uses a traditional HMM employing Baum-Welch EM learning [9]; [15] assigns biological meaning to the latent states and implements a Bayesian HMM via a Metropolis-within-Gibbs algorithm. The number of states must be preset in these two papers. However, this can lead to over- or underfitting if the underlying state structure is not modeled correctly.

To select a fixed number of states in an HMM, [7] investigates the cross-validation method; [25] discusses the three model selection techniques, maximum description length (MDL), maximum *a posteriori* (MAP) estimate and predictive least square (PLS); [30] proposes a Bayesian model merging method. The Bayesian approaches [4], [31], which place nonparametric priors on the number of states, provide alternatives to the model-selection methods discussed in [7], [25], and [30]. The latter method, a nonparametric (countably) infinite-state model that utilizes the hierarchical Dirichlet process (HDP) [31], has proven effective in many applications [13], [19], [22], [31]. To impose state persistence, [10] and [11] propose a *sticky* extension of HDP-HMM, allowing more robust learning of smoothly varying dynamics. Past non-Bayesian work on state duration modeling in HMMs [28] and inhomogeneous HMMs [27], [33] is related to state persistence. Since the traditional Baum-Welch algorithm cannot be used for these methods, effective approximation algorithms for model training and decoding are needed [27], [28], [33].

Motivated by previous work, we develop a simplified form of the sticky HDP-HMM [10], [11], called the sticky hidden Markov model with Dirichlet distribution prior (sticky DD-HMM), and extend the new model structure to analyze array CGH data in all chromosomes for multiple samples. Inference is performed efficiently via a variational Bayesian (VB) analysis [2], [18].

To validate the sticky HMM formulation, we first present example results on synthesized data. We then consider CGH data associated with breast cancer. While there is no explicit “truth” for this problem, the sticky-HMM results based on CGH data

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are compared to complementary results manifested on associated gene-expression data. For that data we employ a completely distinct modeling paradigm, based on factor analysis. Specifically, we consider sparse Bayesian factor analysis [6] and a non-Bayesian penalized matrix decomposition [35]. While these separate analyses on gene-expression data do not explicitly validate our CGH analysis, there is a strong suggestion of biological correspondence.

II. REVIEW OF HIDDEN MARKOV MODEL WITH DIRICHLET PRIORS

A. Hidden Markov Model

The HMM [26] is a generative statistical representation of sequential data, with an underlying discrete Markovian process selecting state-dependent distributions from which observations are drawn. Specifically, for a sequence of length T , an underlying “hidden” state sequence $\mathbf{S} = (s_1, s_2, \dots, s_T)$ is drawn from $p(s_t|s_{t-1}, \dots, s_1) = p(s_t|s_{t-1})$. The observed sequence $\mathbf{X} = (x_1, x_2, \dots, x_T)$ is drawn as $f(\theta_{s_t})$, where $f(\cdot)$ represents the observation model, and θ_{s_t} is the set of parameters for the model indexed by the state at time t , s_t . Note that, given the underlying states, the observations at each time are conditionally independent.

In [26], the number of states associated with an HMM is initialized and fixed. A J -state HMM can be modeled as $\Omega = \{\mathbf{W}, \boldsymbol{\theta}, \mathbf{w}_0\}$, where $\mathbf{W}$ is a $J \times J$ matrix with entry $w_{i,j}$ representing the transition probability from state i to j ; $\boldsymbol{\theta} = \{\theta_1, \theta_2, \dots, \theta_J\}$ where θ_j represents the observation-model parameters associated with state j ; and $\mathbf{w}_0$ is a J -dimensional probability vector defining the probability of being in each of the J states for the first observation.

The data-generating process may be represented as

$$\begin{aligned} x_t &\sim f(\theta_{s_t}); \quad t = 1, \dots, T \\ s_t &\sim \begin{cases} \mathbf{w}_0 = [w_{0,1}, w_{0,2}, \dots, w_{0,J}] & \text{if } t = 1 \\ \mathbf{w}_{s_{t-1}} = [w_{s_{t-1},1}, w_{s_{t-1},2}, \dots, w_{s_{t-1},J}] & \text{if } t > 1. \end{cases} \end{aligned} \quad (1)$$

For given Ω , the joint probability of the observation and the underlying state sequence is expressed as

$$p(\mathbf{X}, \mathbf{S} | \mathbf{W}, \boldsymbol{\theta}, \mathbf{w}_0) = w_{0,s_1} \prod_{t=1}^{T-1} w_{s_t, s_{t+1}} \prod_{t=1}^T p(x_t | \theta_{s_t}). \quad (2)$$

The data likelihood $p(\mathbf{X} | \mathbf{W}, \boldsymbol{\theta}, \mathbf{w}_0)$ can be obtained by integrating over the states using the forward algorithm [26].

B. HMM With Dirichlet Distribution Prior

The priors associated with $\mathbf{w}_0$ and the rows of $\mathbf{W}$ are typically Dirichlet distributions, since these are conjugate to the multinomial likelihood. The standard Dirichlet distribution is written as

$$p(w_1, \dots, w_J | \alpha_1, \dots, \alpha_J) = \frac{\Gamma\left(\sum_{i=1}^J \alpha_i\right)}{\prod_{i=1}^J \Gamma(\alpha_i)} \prod_{i=1}^J w_i^{\alpha_i-1} \quad (3)$$

with the mean and variance of an element, w_j , represented as

$$\mathbb{E}[w_j] = \frac{\alpha_j}{\sum_{i=1}^J \alpha_i}, \quad \mathbb{V}[w_j] = \frac{\alpha_j \left(\sum_{i=1}^J \alpha_i - \alpha_j\right)}{\left(\sum_{i=1}^J \alpha_i\right)^2 \left(\sum_{i=1}^J \alpha_i + 1\right)}. \quad (4)$$

To understand the properties of a draw from the Dirichlet distribution (DD) and to make connections to the infinite HMM with hierarchical Dirichlet process prior (HDP-HMM) discussed in Section II-C, we recall the infinite “stick-breaking” representation [29] for the draw $\mathbf{w} \sim \text{Dir}(\alpha_1, \dots, \alpha_J)$, where $\text{Dir}(\cdot)$ represents the Dirichlet distribution. We define $\alpha = \sum_{i=1}^J \alpha_i$, and $\mathbf{g}_0$ is a base probability vector with j th element $g_{0,j} = \alpha_j/\alpha$. A draw $\mathbf{w} \sim \text{Dir}(\alpha_1, \dots, \alpha_J)$ may be constructed as

$$\begin{aligned} w_j &= \sum_{k=1}^{\infty} h_k \delta(Z_k = j); \quad j = 1, \dots, J \\ h_k &= V_k \prod_{\tau=1}^{k-1} (1 - V_\tau), \quad V_k \sim \text{Beta}(1, \alpha); \\ k &= 1, \dots, \infty \\ Z_k &\sim \mathbf{g}_0; \quad k = 1, \dots, \infty \end{aligned} \quad (5)$$

where $\delta(Z_k = j)$ equals to one if $Z_k = j$, and its zero otherwise; $\text{Beta}(\cdot)$ represents the Beta distribution. Hence, $\mathbf{w}$ is built up as a sum of probability vectors with all zeros and a single randomly selected one (defined by draws from $\mathbf{g}_0$), with these probability vectors multiplied by the stick weights h_k . Note that if α is small then the draws from $\text{Beta}(1, \alpha)$ are such that only a relatively small number of sticks h_k will have significant weight, and hence with high probability a draw $\mathbf{w}$ will only possess a relatively small number of components with significant mass (for large J). To simplify notation below, the infinite-dimensional probability vector $\mathbf{h}$ constructed as above is denoted $\mathbf{h} \sim \text{Stick}(\alpha)$.

C. Infinite HMM With HDP Prior

A draw from a Dirichlet process (DP) may also be represented in stick-breaking form [16], [29]

$$\mathbf{G} = \sum_{j=1}^{\infty} v_j \delta_{\theta_j}, \quad \mathbf{v} \sim \text{Stick}(\gamma), \quad \theta_j \sim H. \quad (6)$$

The advantage of the DP construction, *vis-a-vis* the Dirichlet distribution discussed above, is that in the latter one must explicitly define a truncation level, while the latter in principle may support an infinite set of mixture components (the *truncated* DP is discussed in [16]). The DP is commonly used as a prior on the parameters of a mixture model with unknown number of mixture components [see Fig. 1(a)]. This sampling process is often described via a discrete indicator variable $s_n \sim \mathbf{v}$, indicating which atom generates the n th observation $x_n \sim f(\theta_{s_n})$, for $n = 1, \dots, N$.

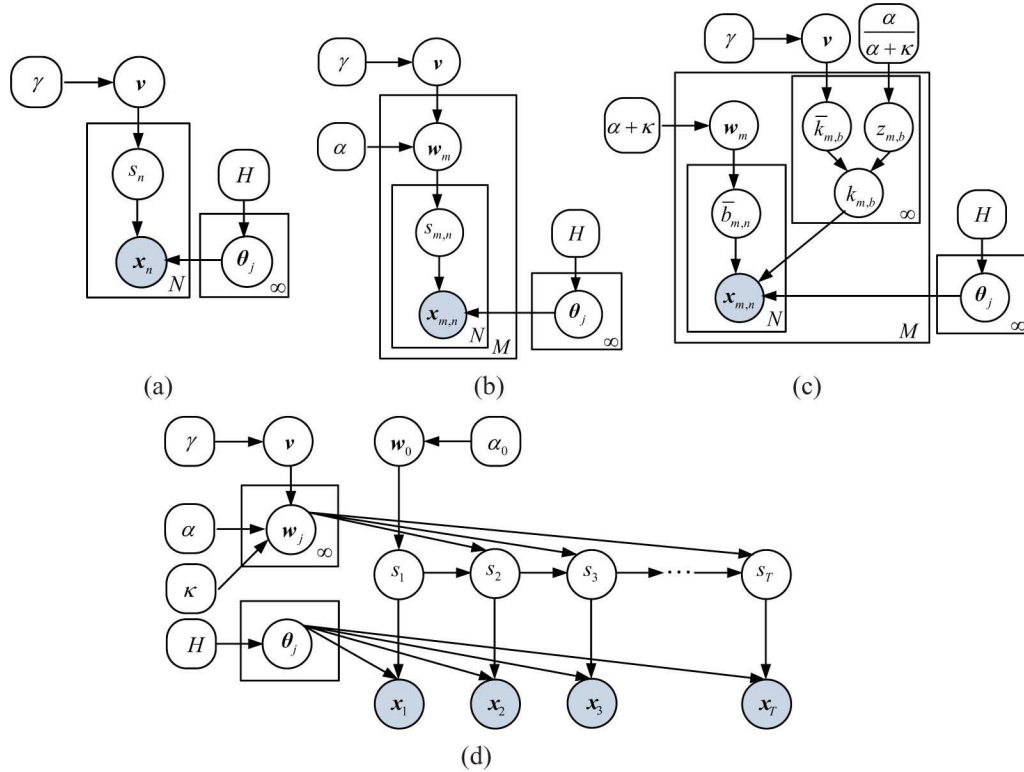


Fig. 1. (a) DPMM in which mixing weight $\mathbf{v} \sim \text{Stick}(\gamma)$ with γ being a hyper-parameter, atom $\theta_j \sim H$ with H denoting the prior distribution for $\{\theta_i\}_{i=1}^{\infty}$, the n th observation's indicator $s_n \sim \mathbf{v}$, and the n th observation $x_n \sim f(\theta_{s_n})$ for $n = 1, \dots, N$. (b) HDP mixture model in which global mixing weight $\mathbf{v} \sim \text{Stick}(\gamma)$ with γ being a hyper-parameter, task-dependent mixing weight $\mathbf{w}_m \sim \text{DP}(\alpha, \mathbf{v})$ with α being a hyper-parameter, atom $\theta_j \sim H$ with H denoting the prior distribution for $\{\theta_i\}_{i=1}^{\infty}$, in the m th task the n th observation's indicator $s_{m,n} \sim \mathbf{w}_m$, and in the m th task the n th observation $x_{m,n} \sim f(\theta_{s_{m,n}})$ for $m = 1, \dots, M$ and $n = 1, \dots, N$. (c) CRF with loyal customers. Customers $x_{m,n}$ sit at table $\bar{b}_{m,n} \sim \mathbf{w}_m$ which considers dish $\bar{k}_{m,b}$, $\bar{z}_{m,b} \sim \mathbf{v}$, but override variables $z_{m,b}$, $\bar{b}_{m,n} \sim \text{Ber}(\alpha/\kappa + \alpha)$ can force the served dish $k_{m,b}$ to be the specialty. The original CRF, as discussed in [31], has $\kappa = 0$ so that $k_{m,b} = \bar{k}_{m,b}$. To allow the degree of self-transition bias to be strongly influenced by data, the gamma prior and beta prior are respectively put on $\alpha + \kappa$ and $\kappa/(\alpha + \kappa)$ with the detailed implementation discussed in [11]. (d) Sticky HDP-HMM where the state evolves as the $t+1$ th observation's indicator $s_{t+1} \sim \mathbf{w}_{s_t}$ with s_t being the state indicator of the t th observation, the state transition probabilities from state j (state-dependent mixing weight) $\mathbf{w}_j \sim \text{DP}(\alpha + \kappa, (\alpha \mathbf{v} + \kappa \delta_j)/(\alpha + \kappa))$ with the detailed implementation discussed in [11] for $j = 1, \dots, \infty$, global mixing weight $\mathbf{v} \sim \text{Stick}(\gamma)$ with γ being a hyper-parameter, and the t th observation $x_t \sim f(\theta_{s_t})$ for $t = 1, \dots, T$. The original HDP-HMM has $\kappa = 0$.

The HDP [31] extends the DP to cases in which groups of data are produced by related, yet unique, generative processes. In the HDP structure, the base probability measure, G_0 , is itself drawn from a Dirichlet process. The formal notation is as follows:

$$G_m \sim \text{DP}(\alpha, G_0), \quad G_0 \sim \text{DP}(\gamma, H) \quad (7)$$

where G_m represents the prior distribution associated with group m . The HDP is a two-level model, where the distribution on the atoms is shifted from the continuous H to the discrete (but countably infinite) G_0 . An alternative representation of the model is

$$G_m = \sum_{j=1}^{\infty} w_{m,j} \delta_{\theta_j}, \quad \mathbf{w}_m \sim \text{DP}(\alpha, \mathbf{v}),$$

$$G_0 = \sum_{j=1}^{\infty} v_j \delta_{\theta_j}, \quad \mathbf{v} \sim \text{Stick}(\gamma), \quad \theta_j \sim H. \quad (8)$$

Observation $x_{m,n}$ is associated with one of the global set of discrete parameters via an indicator random variable $s_{m,n} \sim \mathbf{w}_m$; see Fig. 1(b).

The HDP can be used to develop an HMM with an unknown number of states [31]. For this HDP-HMM, each HDP group-

specific distribution, $\mathbf{w}_m$, is a state-specific transition distribution and, due to the infinite state space, there are infinitely many groups. Let s_t denote the state of the Markov chain at time t . As is done typically, $s_1 \sim \mathbf{w}_0$. For $t > 1$, we have the Markov process $s_t \sim \mathbf{w}_{s_{t-1}}$, so that s_{t-1} indexes the group to which x_t is assigned. The current HMM state s_t then indexes the parameter θ_{s_t} used to generate observation x_t [see Fig. 1(d)]. According to (8), the HDP formulation effectively selects the number of states and their observation parameters via the top-level DP and uses the mixing weights as the prior for a second-level Dirichlet distribution from which the transition probabilities are drawn. Importantly, since G_0 is composed of a discrete set of atoms, the state-dependent probabilities are shared across the different $\mathbf{w}_m$. The lack of conjugacy between the two levels in the model, however, means that a variational solution [2], [18] is difficult. The infinite HDP-HMM has been implemented by a Gibbs sampler and a slice sampler separately in [11] and [13], of which the first method is a truncated approximation.

D. Sticky HMM With HDP Prior

In the HDP-HMM construction, by sampling $\mathbf{w}_j \sim \text{DP}(\alpha, \mathbf{v})$, the HDP encourages states to have a similar

transition distribution ($\mathbb{E}[w_{j,i}] = v_i$). However, it does not differentiate self-transitions from moves between states. In many applications, one would like to be able to incorporate prior knowledge that slow, smoothly varying dynamics are probable (i.e., that it is likely to stay in the same state for prolonged time periods). When modeling systems with state persistence, traditional HMM design may lead to many redundant (essentially duplicate) states into which transitions occur, with the effect of manifesting a persistence in the observation statistics. However, such models impede our ability to identify a single dynamical model which best explains the observations (it undermines interpretability). Therefore, [10] and [11] proposed to instead sample transition distributions w_j as follows:

$$w_j \sim \text{DP} \left(\alpha + \kappa, \frac{\alpha v + \kappa \delta_j}{\alpha + \kappa} \right). \quad (9)$$

Here, $(\alpha v + \kappa \delta_j)$ indicates that an amount $\kappa > 0$ is added to the j th component of αv . When $\kappa = 0$ the original HDP-HMM is recovered. Because positive κ values increase the prior probability $\mathbb{E}(w_{j,j})$ of self-transitions, [10], [11] referred this extension as the *sticky* HDP-HMM (the model favors “sticking” in the same state for prolonged periods).

The inference algorithm is simplified by using Chinese restaurant franchise (CRF) with loyal customers discussed in [10] and [11] if we introduce the auxiliary random variables $k_{j,b}$ and $z_{j,b}$ as follows:

$$z_{j,b} \sim \text{Ber} \left(\frac{\alpha}{\alpha + \kappa} \right), \quad k_{j,b} \sim \begin{cases} \delta_j & \text{if } z_{j,b} = 0 \\ v & \text{if } z_{j,b} = 1. \end{cases} \quad (10)$$

where $\text{Ber}(\cdot)$ represents the Bernoulli distribution, $k_{j,b}$ represents the *served* dish of table b . As in Fig. 1(c), if $k_{j,b} \sim v$ is used to represent the *considered* dish without taking the restaurant’s specialty into consideration, when $\kappa = 0$, since $z_{j,b} = 1$ for all tables, the considered dish is always the served dish, i.e., $k_{j,b} = \bar{k}_{j,b}$, which is directly drawn from v (in this way, the original HDP-HMM is recovered); when $\kappa \neq 0$, with some probability, this considered dish is overridden and the table is served the specialty dish j (in this way, the stickiness is imposed). In practice, the gamma prior and beta prior are respectively put on $\alpha + \kappa$ and $\kappa/(\alpha + \kappa)$, which allows the degree of self-transition bias to be strongly influenced by the statistics of observed data, as desired. Due to the HDP structure, the proposed sticky model was implemented by a Gibbs sampler in [10] and [11], which is a truncated approximation to the infinite sticky HDP-HMM. However, such MCMC [14] inference may be impractical computationally when considering a large dataset of sequential data.

III. STICKY HMM WITH DIRICHLET DISTRIBUTION PRIOR

A. Model Construction

We seek a simplified implementation of the sticky HDP-HMM. Specifically, the proposed model, termed the sticky DD-HMM, is represented as

$$x_t \sim f(\theta_{s_t}); \quad t = 1, \dots, T$$

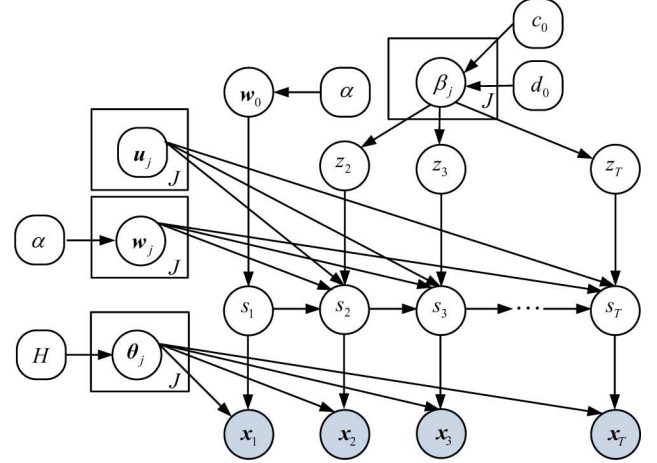


Fig. 2. Graph of the sticky DD-HMM. The detailed generative process is described in (11).

$$\begin{aligned} s_t &\sim \begin{cases} w_0 & \text{if } t = 1 \\ u_{s_{t-1}} & \text{if } t > 1 \text{ and } z_t = 0 \\ w_{s_{t-1}} & \text{if } t > 1 \text{ and } z_t = 1; \end{cases} \quad t = 1, \dots, T \\ z_t &\sim \text{Ber}(\beta_{s_{t-1}}); \quad t = 2, \dots, T \\ \beta_j &\sim \text{Beta}(c_0, d_0); \quad j = 1, \dots, J \\ w_0 &\sim \text{Dir} \left(\frac{\alpha}{J}, \dots, \frac{\alpha}{J} \right) \\ w_j &\sim \text{Dir} \left(\frac{\alpha}{J}, \dots, \frac{\alpha}{J} \right); \quad j = 1, \dots, J \\ [u_1, \dots, u_J] &= I_J \\ \theta_j &\sim H; \quad j = 1, \dots, J \end{aligned} \quad (11)$$

where I_J denotes a $J \times J$ identity matrix, so each u_j with all zeros but a one in the j th component. A graphical representation of this model is depicted in Fig. 2.

For future notational convenience, below we represent

$$\begin{aligned} s_t &\sim \begin{cases} w_0 & \text{if } t = 1 \\ u_{s_{t-1}} & \text{if } t > 1 \text{ and } z_t = 0 \\ w_{s_{t-1}} & \text{if } t > 1 \text{ and } z_t = 1; \end{cases} \quad t = 1, \dots, T \\ [u_1, \dots, u_J] &= I_J \\ z_t &\sim \text{Ber}(\beta_{s_{t-1}}); \quad t = 2, \dots, T \end{aligned} \quad (12)$$

as $s_t \sim \text{S-HMM}(\{w_j, \beta_j\}_{j=1}^J, w_0, s_{t-1})$.

Recall from (5) that draws $w \sim \text{Dir}(\alpha/J, \dots, \alpha/J)$ will have components with an appreciable number of components defined by $\text{Stick}(\alpha)$, and hence a relatively small α setting will allow inference on which subset of J possible states are actually needed based on the data (J may be set large).

In (11), the parameter β_j controls the degree of “stickiness” for state j (i.e., the probability of staying in state j), and this is inferred by the data (with a beta prior imposed). The state-dependent β_j therefore plays the role of $\alpha/(\alpha + \kappa)$ in (10), and can be different for different states in this model. Further, the u_j in (11) play the role of the delta function in (10). We refer to this as an approximation to a point measure—delta function—at point j . Therefore, the model in (11) has many of the characteristics of the original sticky HDP-HMM, but it yields simplified inference, as discussed further below.

B. Relationship Between Sticky DD-HMM and Sticky HDP-HMM

Comparing the sticky HDP-HMM in [10] and [11] to the proposed sticky DD-HMM, there are two main modifications: i) using the DD prior to replace the HDP prior and ii) using a special discrete distribution to approximate the point measure δ_j in (10). Since ii) is just a direct and small modification, we will mainly discuss i) in the following.

In the HDP structure (8), the draw $\mathbf{w}_j \sim \text{DP}(\alpha, \mathbf{v})$ may be presented in stick-breaking form, with the i th element of $\mathbf{w}_j$ construct as $w_{j,i} = \sum_{k=1}^{\infty} h_{j,k} \delta(Z_{j,k} = i)$, with $\mathbf{h}_j \sim \text{Stick}(\alpha)$ and $Z_{j,k} \sim \mathbf{v}$. We may also truncate the draw $\mathbf{v} \sim \text{Stick}(\gamma)$ to J sticks (denoted $\mathbf{v}_J \sim \text{Stick}_J(\gamma)$), for large J [16]. Note that for the HMM transition matrix $\mathbf{W}$ is square, and therefore we consider J rows. Using these representations, the truncated HDP construction in HDP-HMM may be represented as

$$\begin{aligned} w_{j,i} &= \sum_{k=1}^{\infty} h_{j,k} \delta(Z_{j,k} = i); \quad i = 1, \dots, J; \quad j = 1, \dots, J \\ \mathbf{h}_j &\sim \text{Stick}(\alpha); \quad j = 1, \dots, J \\ Z_{j,k} &\sim \mathbf{v}_J; \quad k = 1, \dots, \infty; \quad j = 1, \dots, J \\ \mathbf{v}_J &\sim \text{Stick}_J(\gamma). \end{aligned} \quad (13)$$

Note that we truncate $\text{Stick}(\gamma)$ to J sticks [16], but do *not* truncate $\text{Stick}(\alpha)$. In other words, although the transition matrix $\mathbf{W}$ is truncated, the $\mathbf{h}$ vector is still an infinite vector.

By comparing (13)–(5) we observe that the difference between the truncated HDP-HMM (with truncation only for $\text{Stick}_J(\gamma)$) is that in the HDP-HMM the atoms are drawn from $\mathbf{v}_J$, which is drawn from $\text{Stick}_J(\gamma)$; in the DD-HMM, $\mathbf{v}_J$ is fixed as $(1/J, \dots, 1/J)$. Thus, the HDP construction imposes that certain states are consistently more likely to be used globally (those with large components in $\mathbf{v}_J$), and this information may be important for problems in which there is limited data. In [11], the authors show a specific example for which using the HDP construction is of particular importance. Nevertheless, in our experiments for the type of problems of interest here, the DD-HMM construction has performed well, and the computational analysis of this model is relatively straightforward, as discussed in Section IV.

Setting a large J does not imply that we believe that there are actually J states, since from (5) only a relatively small set of components in $\mathbf{w}_j$ will have appreciable amplitude (the same type of motivation for the stick-breaking view of DP and HDP). We also emphasize that the stick-breaking representation of a draw from a Dirichlet distribution has been introduced above to make the connection between the proposed model and a truncated representation of HDP-HMM. However, when actually performing inference, it is often simpler to just draw directly from $\text{Dir}(\alpha/J, \dots, \alpha/J)$.

IV. VARIATIONAL BAYESIAN INFERENCE

The proposed sticky DD-HMM model may be readily implemented via MCMC sampling. However, we employ variational Bayesian (VB) [2], [18] inference in this paper, as a compromise between accuracy and efficiency for large-scale problems (the detailed reason for selecting VB will be discussed further

in Section VI-B). One may in principle also employ a VB analysis to the HDP-based construction in [10] and [11], using techniques like those considered in [19].

Bayesian inference seeks to estimate the posterior distribution of the latent variables Ψ , given the observed data $\mathbf{X}$ and hyperparameters Υ :

$$p(\Psi | \mathbf{X}, \Upsilon) = \frac{p(\mathbf{X} | \Psi, \Upsilon) p(\Psi | \Upsilon)}{\int p(\mathbf{X} | \Psi, \Upsilon) p(\Psi | \Upsilon) d\Psi} \quad (14)$$

where the denominator $\int p(\mathbf{X} | \Psi, \Upsilon) p(\Psi | \Upsilon) d\Psi = p(\mathbf{X} | \Upsilon)$ is the model evidence (marginal likelihood). VB inference seeks a variational distribution $q(\Psi)$ to approximate the true posterior distribution of the latent variables $p(\Psi | \mathbf{X}, \Upsilon)$. The expression

$$\log p(\mathbf{X} | \Upsilon) = \mathcal{L}(q(\Psi)) + \mathcal{KL}(q(\Psi) \parallel p(\Psi | \mathbf{X}, \Upsilon)) \quad (15)$$

with

$$\mathcal{L}(q(\Psi)) = \int q(\Psi) \log \frac{p(\mathbf{X} | \Psi, \Upsilon) p(\Psi | \Upsilon)}{q(\Psi)} d\Psi \quad (16)$$

forms a lower bound for $\log p(\mathbf{X} | \Upsilon)$ so that $\log p(\mathbf{X} | \Upsilon) \geq \mathcal{L}(q(\Psi))$, since $\mathcal{KL}(q(\Psi) \parallel p(\Psi | \mathbf{X}, \Upsilon)) \geq 0$. Accordingly, the goal of minimizing the KL divergence between the variational distribution and the true posterior reduces to adjusting $q(\Psi)$ to maximize (16).

VB inference [2], [18] assumes a factorized $q(\Psi)$, i.e., $q(\Psi) = \prod_k q_k(\Psi_k)$, typically with the same form as employed in $p(\Psi | \mathbf{X}, \Upsilon)$. With such an assumption, the variational distributions can be updated iteratively to increase the lower bound $\mathcal{L}(q(\Psi))$. The mean-field variational distribution for the proposed sticky DD-HMM model is

$$\begin{aligned} q(\Psi) &= q(\mathbf{w}_0) \left[\prod_{j=1}^J q(\mathbf{w}_j) \right] q(\beta) \left[\prod_{t=1}^T q(s_t) \right] \\ &\quad \times \left[\prod_{t=2}^T q(z_t) \right] \left[\prod_{j=1}^J q(\theta_j) \right] \end{aligned} \quad (17)$$

where $\{\{\mathbf{w}_i\}_{i=0}^J, \beta, \{s_t\}_{t=1}^T, \{z_t\}_{t=2}^T, \{\theta_j\}_{j=1}^J\}$ are the latent variables in this model.

A general method for performing variational inference for conjugate-exponential Bayesian networks outlined in [34] is as follows: For a given node in a graphic model, write out the posterior as though everything were known, take the logarithm, the expectation with respect to all unknown parameters and exponentiate the result. The variational lower bound $\mathcal{L}(q(\Psi))$, which is a function of variational distributions $q(\Psi)$, is optimized by iteratively taking derivatives with respect to the hyper-parameters for each term obtained above, which is associative with a node in the graphic model, and setting the result to zero while fixing the hyper-parameters of the other terms. Within each iteration, the lower bound is increased until the model converges. In the following experiments, we terminate the algorithm when the change in $\mathcal{L}(q(\Psi))$ is negligibly small. Since it requires computational resources comparable to the expectation-maximization (EM) algorithm, variational inference is fast relative to MCMC methods [14].

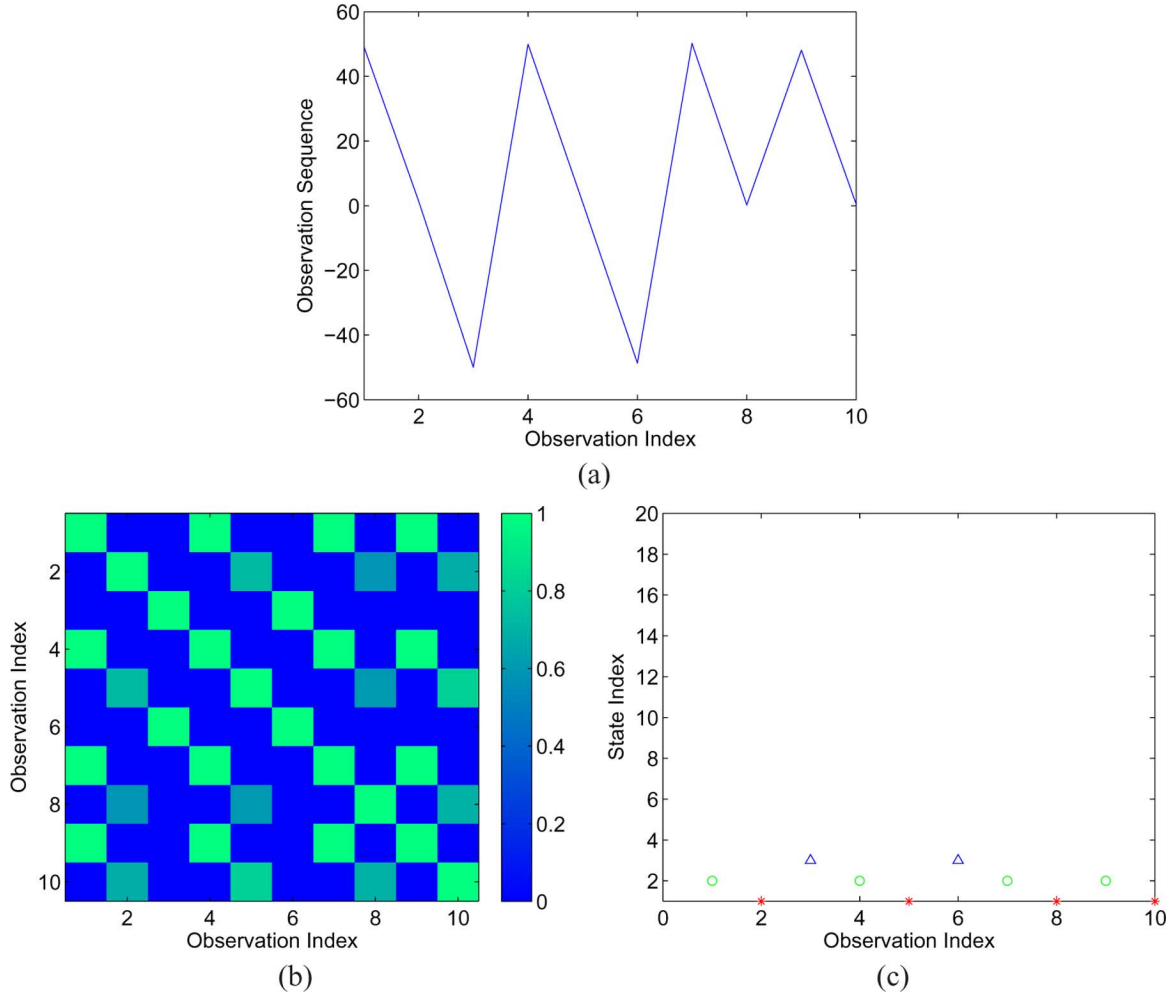


Fig. 3. Synthetic example I. (a) Generated observation sequence, (b) HDP-HMM using MCMC inference [11], (c) DD-HMM using VB inference. (b) denotes the fraction of times within the collection samples that a given portion of the observation sequence shares the same underlying state. In (c), the colored symbols denote ground truth: red * represents the state with $\mu_1 = 0$, green o represents the state with $\mu_2 = 50$, blue $\triangle$ represents the state with $\mu_3 = -50$; and each observation is assigned to a state index ($J = 20$), with the index shown along the vertical axis.

The detailed derivation of the update equations for the variational posteriors is summarized in the Appendix. The corresponding lower bound can be computed by substituting the updated $q(\Psi)$ and the prior distributions $p(\Psi|\mathbf{Y})$ into (16), the detailed derivation of which is omitted for brevity. Like other optimization algorithms, VB inference may converge to a local-optimal solution. However, such a problem can be alleviated by running the algorithm multiple times from different initializations and then using the solution that maximizes the variational model evidence.

V. EXPERIMENTS ON SYNTHETIC DATA

Here we present experimental results on synthetic data. In the following comparisons we used the non-sticky and sticky HDP-HMM implementations discussed in [11] and available online.¹ The HDP-HMM hyper-parameters are set as the defaults, and the truncation level is set to be the same as that of our DD-HMM. The MCMC results obtained using the HDP-HMM software [11] were computed using 5000 burn-in iterations and 5000 collection iterations.

A. DD-HMM vs HDP-HMM

We synthesized data from the following HMM

$$\begin{aligned}
 x_t &\sim \text{Norm}(\mu_{s_t}, \sigma_{s_t}^{-1}), \quad \mu_1 = 0, \quad \mu_2 = 50, \\
 \mu_3 &= -50, \quad \sigma_1^{-1} = \sigma_2^{-1} = \sigma_3^{-1} = 1 \\
 \mathbf{W} &= \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{bmatrix}, \quad \mathbf{w}_0 = [\frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}]. \quad (18)
 \end{aligned}$$

The data can be generated according to (1). Here we generated a sequence of length $T = 10$ from this model. The generated observation sequence is shown in Fig. 3(a). Please note the state transition matrix $\mathbf{W}$ assigns zero probability to self-transitions in this example.

Since the data likelihood is a univariate Gaussian distribution, we put corresponding Gaussian and Gamma priors on the mean and precision parameters. To apply the DD-HMM and HDP-HMM models on the generated data, we set the truncation level as $J = 20$. In DD-HMM, as discussed in Section II-B, to uncover the number of states, we place sparseness-promoting priors $\text{Dir}([10^{-3}, \dots, 10^{-3}])$ on each $\mathbf{w}_j$ with $j = 0, \dots, J$,

¹<http://www.mit.edu/ebfox/software.html>

which means $\alpha/J = 10^{-3}$ in (11). All the DD-HMM hyper-parameters have not been optimized or tuned, and when presenting examples on real data below, we examine model sensitivity to setting J and α .

When employing MCMC inference, atoms vary across the collected samples from the posterior [17], therefore, we cannot get an overall state label decision based on all collected samples; using VB inference, atoms are fixed in the posterior computation, and we obtain a posterior distribution on s_t , i.e., $\langle s_{t,j} \rangle$ for $j = 1, \dots, J$, and approximate the membership for each measurement by assigning it to the state with largest probability. Therefore, besides the relatively fast computation, another advantage of VB inference is the avoidance of the “label-switching” problem associated with MCMC [17]. Fig. 3(c) presents the “hard” (most likely) decisions employed to provide state labels (the Bayesian analysis can also yield a “soft” decision in terms of a full posterior distribution) for the generated sequential data via DD-HMM or sticky DD-HMM using VB inference. To indicate the ground truth, different symbols and colors are used to represent different states in the generated observation. In Fig. 3(b) is shown the fraction of times within the collection samples that a given portion of the signal share the same underlying state. By contrast, from the results in Fig. 3(b) and (c), both HDP-HMM and DD-HMM are in agreement with ground truth, which infer a proper number of states and correct membership of each observation.

B. Sticky DD-HMM vs Sticky HDP-HMM

We synthesized data from the following sticky HMM

$$\begin{aligned} x_t &\sim \text{Norm}(\mu_{s_t}, \sigma_{s_t}^{-1}), \quad \mu_1 = 0, \quad \mu_2 = 50, \\ \mu_3 &= -50, \quad \sigma_1^{-1} = \sigma_2^{-1} = \sigma_3^{-1} = 100 \\ \mathbf{W} &= \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.50 & 0.25 \\ 0.25 & 0.25 & 0.50 \end{bmatrix}, \quad \mathbf{w}_0 = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}, \\ \boldsymbol{\beta} &= [0.01 \quad 0.05 \quad 0.03]. \end{aligned} \quad (19)$$

The data can be generated according to (12). From this model we generated a sequence of length $T = 400$. The generated observation sequence is shown in Fig. 4(a). Please note strong temporal persistence is captured by both the $\boldsymbol{\beta}$ parameter and the state transition matrix $\mathbf{W}$ in this example.

Here we set the truncation level as $J = 20$. The hyper-parameters of the non-sticky DD-HMM are set the same as those used in Section V-A. In the sticky DD-HMM, to encourage the state persistence, we place Beta (0.1,0.9) priors on each β_j ; priors on $\mathbf{w}_0$ and each $\mathbf{w}_j$ are set Dir($[10^{-3}, \dots, 10^{-3}]$), as in the non-sticky DD-HMM in Section V-A, for $j = 1, \dots, J$. All the above hyper-parameters of our model have not been optimized or tuned.

As demonstrated in Fig. 4(b) and (d), without an extra self-transition bias, the original (non-sticky) HDP-HMM or DD-HMM rapidly transitions among redundant states. By contrast, from the results in Fig. 4(c) and (e), both sticky HMMs are in agreement with ground truth, inferring a proper number of states and correct membership of each observation.

VI. CGH DATA ANALYSIS APPLICATION

A. Multi-Task Learning for CGH Data

For the motivating CGH problem of interest here, one typically has access to different persons’ CGH sequence data from all 23 chromosomes. We wish to learn a sticky HMM for each of these chromosomes, and recognize that there is likely statistical inter-relationships between the chromosomes that may be exploited. We therefore wish to learn sticky HMMs for data sets $m = 1, \dots, M$, for $M = 23$, and the data set from chromosome m , which is composed of L measurement sequences from different persons, is termed learning task m . The learning of sticky HMMs for all $M = 23$ tasks jointly is referred to as multi-task learning [5]. We here extend the discussion in the previous sections to sticky HMM learning in a multi-task setting. Similar MTL techniques have been successfully applied to information retrieval [3] and computer vision [32], as well as music (sequential data) analysis [22].

Based on the biological insights from [15], the different HMM states, related to copy number, are linked to health and disease. Where these states are manifested across the 23 chromosomes impacts the disease particulars (since where on the chromosome the states are manifested is linked to genes of importance for the disease; each gene resides on a specific chromosome). Since the state-transition statistics impact the probability of manifesting particular states, we here assume these transition probabilities are chromosome-dependent (the states will not have the same probability of being manifested across the 23 chromosomes, as a given disease will locate anomalous states in specific regions of the chromosomes). However, the observation statistics of healthy or unhealthy copy-number rate is assumed to *not* depend on which specific chromosome is being considered [15]; i.e., we assume that the observation statistics of anomalous or typical copy-rates does not define the disease, *where* across the 23 chromosomes these states exist defines the particulars of disease. So motivated, in the proposed MTL model, each of the $M = 23$ chromosomes is assumed to have unique state-transition statistics, but the state-dependent observation statistics are shared across the M tasks. This implies that the different chromosomes share the same underlying states, but the state-dependent transition probabilities are chromosome dependent. The state-dependent observations for the CGH data are assumed to be drawn from a Gaussian model [15], and the overall model is summarized as (20), shown at the bottom of the next page, where $x_t^{(m,l)}$ denotes the l th sample’s CGH measurement at position t in chromosome m , and $s_t^{(m,l)}$ is the corresponding state indicator; $\text{Norm}(\cdot)$ represents the Normal distribution, and $\text{Ga}(\cdot)$ represents the Gamma distribution. Here Normal-Gamma prior (as is $H(\cdot)$) is conjugate to the Normal likelihood (as is $f(\cdot)$).

Each chromosome has its own state-transition statistics, defined by $\mathbf{w}_0^{(m)}$, $\{\mathbf{w}_j^{(m)}\}_{j=1,J}$ and $\{\mathbf{u}_j^{(m)}\}_{j=1,J}$ with chromosome $m \in \{1, \dots, 23\}$. The state-dependent “stickiness,” defined by β_j for state j , is shared across the 23 chromosomes, as are the observation statistics defined by (μ_j, σ_j) . Since $\mu_1 \sim \delta_0$,

state $j = 1$ has an imposed mean of zero, and this corresponds to the no/low copy number state (normal state). In this model, we learn DD-HMMs with independent state transition matrixes for each of the tasks (chromosomes) as well as share the same state set across all tasks.

For the MTL sticky DD-HMM, a similar inference technique to that discussed in Section IV can be easily extended from STL to MTL. Here we omitted those details for brevity.

B. Experimental Results on CGH Data

We examine the performance of the sticky DD-HMM on a breast cancer data set described in [8] and available online.² The breast cancer data is composed of $L = 89$ tissue samples for which both array CGH and gene expression measurements are available. There are in total 22215 gene-expression measurements and 2149 CGH measurements. We apply the proposed MTL sticky DD-HMM to analyze the CGH data, and make comparisons to the MTL original (non-sticky) DD-HMM and the STL Bayesian HMM (BHMM) proposed in [15], for which the codes are included in the Bioinformatics toolbox 3.3 of Matlab software.³ As a comparison/validation of the CGH data, the corresponding gene expression data are analyzed via two factor-analysis formulations: sparse Bayesian factor analysis (SBFA) [6] and the related (non-Bayesian) penalized matrix decomposition (PMD) method [35]. All Bayesian results expect those of BHMM are presented using VB inference; for the size of this problem (with data from all chromosomes analyzed jointly, namely multi-task learning) MCMC inference is very expensive computationally.

In the following experiments, we set the state truncation level to $J = 20$ (similar results were found for larger truncations). The hyperparameters of sticky DD-HMM are set as in Section V-B. All parameters of Bayesian HMM are set as the defaults.

The proposed model learns the posterior state-transition matrixes of each chromosome, and in so doing infers an estimate

of the proper number of states (using the multi-task framework of Section VI-A. If state i is used for chromosome m , then the posterior probability on this state usage, $\sum_{j=1}^J \langle w_{j,i}^{(m)} \rangle$ should be nonzero, where $\langle w_{j,i}^{(m)} \rangle$ denotes the posterior expectation of state transition probability from state j to state i for chromosome m . As shown in Fig. 5, although we initialized the truncation level to $J = 20$, for the CGH data only four states (state 1, state 2, state 3 and state 5) are inferred for all 23 chromosomes, since only the posterior usage probabilities on each of these four states are nonzero. As indicated in the sticky DD-HMM described in (20), for the first state $\mu_1 = 0$ (corresponding to no or a low level of copying); the inferred posterior means of μ_i for $i = 2, 3, 5$ are $\tilde{r}_2 = -0.3709$, $\tilde{r}_3 = 0.4674$ and $\tilde{r}_5 = 1.1709$. The second state corresponds to a copy-number reduction state, the third state corresponds to a small copy-number increase, and the fifth state corresponds to marked copy-number amplification. By contrast, the original (non-sticky) DD-HMM, without a constraint on the first state and without stickiness, inferred 9 states, of which the posterior means are $\tilde{r}_1 = 1.3810$, $\tilde{r}_2 = -0.5696$, $\tilde{r}_3 = -0.0163$, $\tilde{r}_4 = 0.4380$, $\tilde{r}_5 = 0.1891$, $\tilde{r}_6 = -0.2854$, $\tilde{r}_7 = 1.0426$, $\tilde{r}_8 = 0.7373$ and $\tilde{r}_9 = 0.0081$. Thus, two states correspond to normal states, two states correspond to copy number loss state, three correspond to small copy number gain state, and two correspond to multiple copy gains or amplifications state, as computed via DD-HMM (constituting the state redundancy required to manifest stickiness in the observation statistics). The *inferred* four states of the sticky-HMM is consistent with the biologically motivated and *imposed* four states employed in [15] (we also note that when we analyzed each chromosome one at a time, rather than employing MTL, we obtained results very similar to those in [15]).

In CGH data analysis, we desire the assignment of state labels to each CGH fragment, and wish to detect the copy loss or gain based on the state labels. Here we employed the Bayesian HMM [15], DD-HMM and sticky DD-HMM to compare state labeling performance. Due to “label-switching” problem associated with MCMC [17], we didn’t use non-sticky and sticky HDP-HMMs using MCMC. In [15] the authors explicitly imposed four

$$\begin{aligned}
 x_t^{(m,l)} &\sim \text{Norm}\left(\mu_{s_t^{(m,l)}}, \sigma_{s_t^{(m,l)}}^{-1}\right); \quad t = 1, \dots, T_m; \\
 m &= 1, \dots, 23; \quad l = 1, \dots, L \\
 s_t^{(m,l)} &\sim \text{S-HMM}\left(\left\{\mathbf{w}_j^{(m)}, \beta_j\right\}_{j=1}^J, \mathbf{w}_0^{(m)}, s_{t-1}^{(m,l)}\right); \\
 t &= 1, \dots, T_m; \quad m = 1, \dots, 23; \quad l = 1, \dots, L \\
 \mathbf{w}_0^{(m)} &\sim \text{Dir}\left(\frac{\alpha}{J}, \dots, \frac{\alpha}{J}\right); \quad m = 1, \dots, 23 \\
 \mathbf{w}_j^{(m)} &\sim \text{Dir}\left(\frac{\alpha}{J}, \dots, \frac{\alpha}{J}\right); \quad j = 1, \dots, J; \\
 m &= 1, \dots, 23 \\
 \beta_j &\sim \text{Beta}(c_0, d_0); \quad j = 1, \dots, J \\
 (\mu_j, \sigma_j) &\sim \begin{cases} \delta_0 \times \text{Ga}(b^{(0)}, \lambda^{(0)}) & \text{if } j = 1 \\ \text{Norm}\left(r^{(1)}, \frac{1}{(t^{(1)}\sigma_j)}\right) - \text{Ga}(b^{(1)}, \lambda^{(1)}) & \text{if } j > 1; \quad j = 1, \dots, J \end{cases} \quad (20)
 \end{aligned}$$

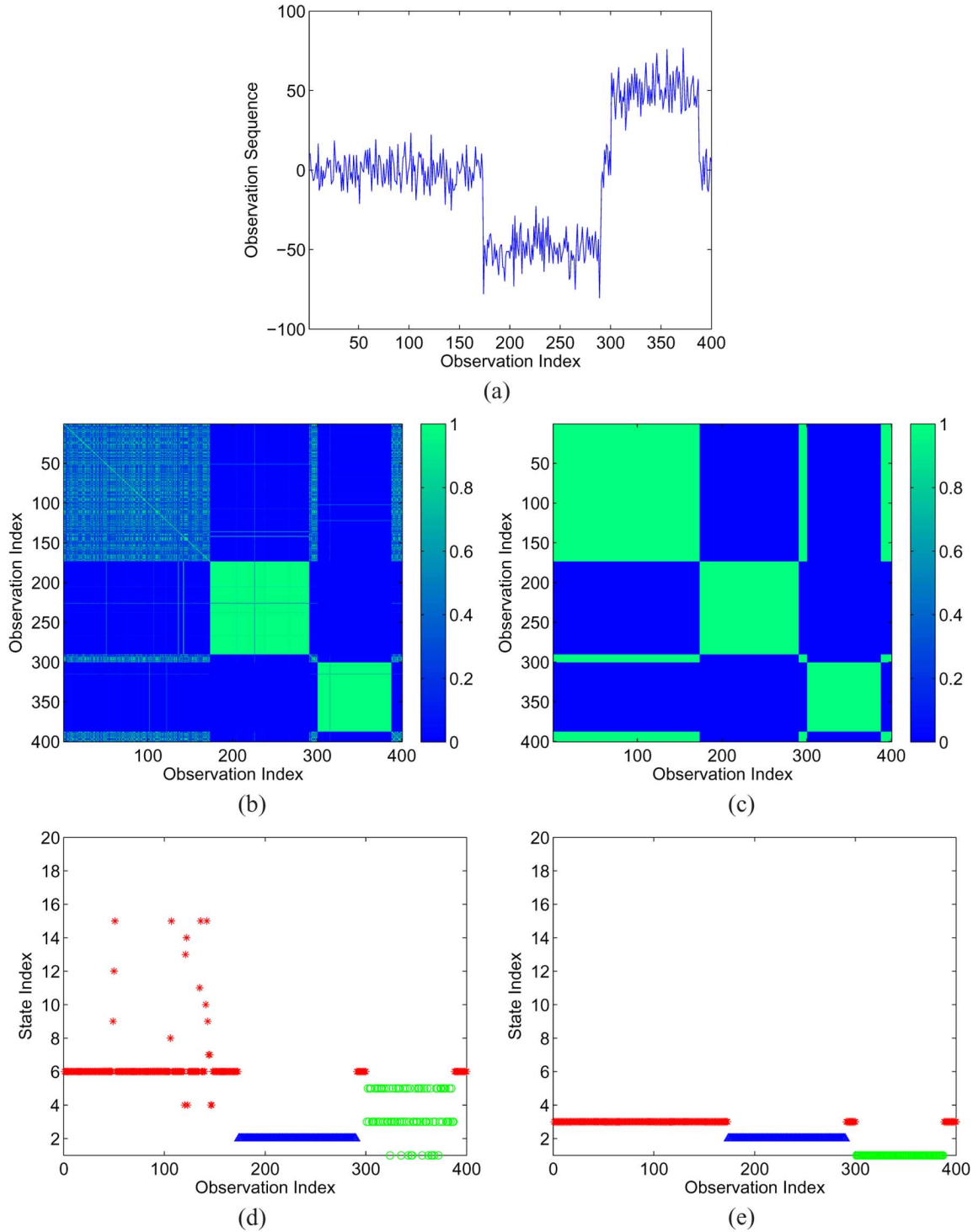


Fig. 4. Synthetic example II. (a) Generated observation sequence, (b) HDP-HMM using MCMC inference [11], (c) sticky HDP-HMM using MCMC inference [11], (d) DD-HMM using VB inference, (e) sticky DD-HMM using VB inference. (b) and (c) denote the fraction of times within the collection samples that a given portion of the observation sequence shares the same underlying state. In (d) and (e), the colored symbols denote ground truth: red * represents the state with $\mu_1 = 0$, green o represents the state with $\mu_2 = 50$, blue Δ represents the state with $\mu_3 = -50$; and each observation is assigned to a state index ($J = 20$), with the index shown along the vertical axis.

states, for which the mean parameter μ_i for $i = 1, 2, 3, 4$ has the constraint $\mu_1 < \mu_2 < \mu_3 < \mu_4$. The priors for the means are: $\mu_1 \sim \text{Norm}(-1, \tau_1^2) \cdot \delta(\mu_1 < -\epsilon)$, $\mu_2 \sim \text{Norm}(0, \tau_2^2) \cdot \delta(-\epsilon < \mu_2 < \epsilon)$, $\mu_3 \sim \text{Norm}(0.58, \tau_3^2) \cdot \delta(\epsilon < \mu_3 < 0.58)$, and $[\mu_4 | \mu_3, \sigma_3^{-1}] \sim \text{Norm}(1, \tau_4^2) \cdot \delta(< \mu_4 > \mu_3 + 3\sigma_3)$, where the parameter ϵ needs to be set in [15], and the results

may be sensitive to how ϵ is set. Such a prior structure also introduces some difficulties for inference, and the computationally expensive Metropolis-Hastings (MH) method [14] is used within Gibbs sampling [15]. In addition, [15] learns independent HMMs for the CGH data in each chromosome separately; the MTL approach developed in Section VI-A was

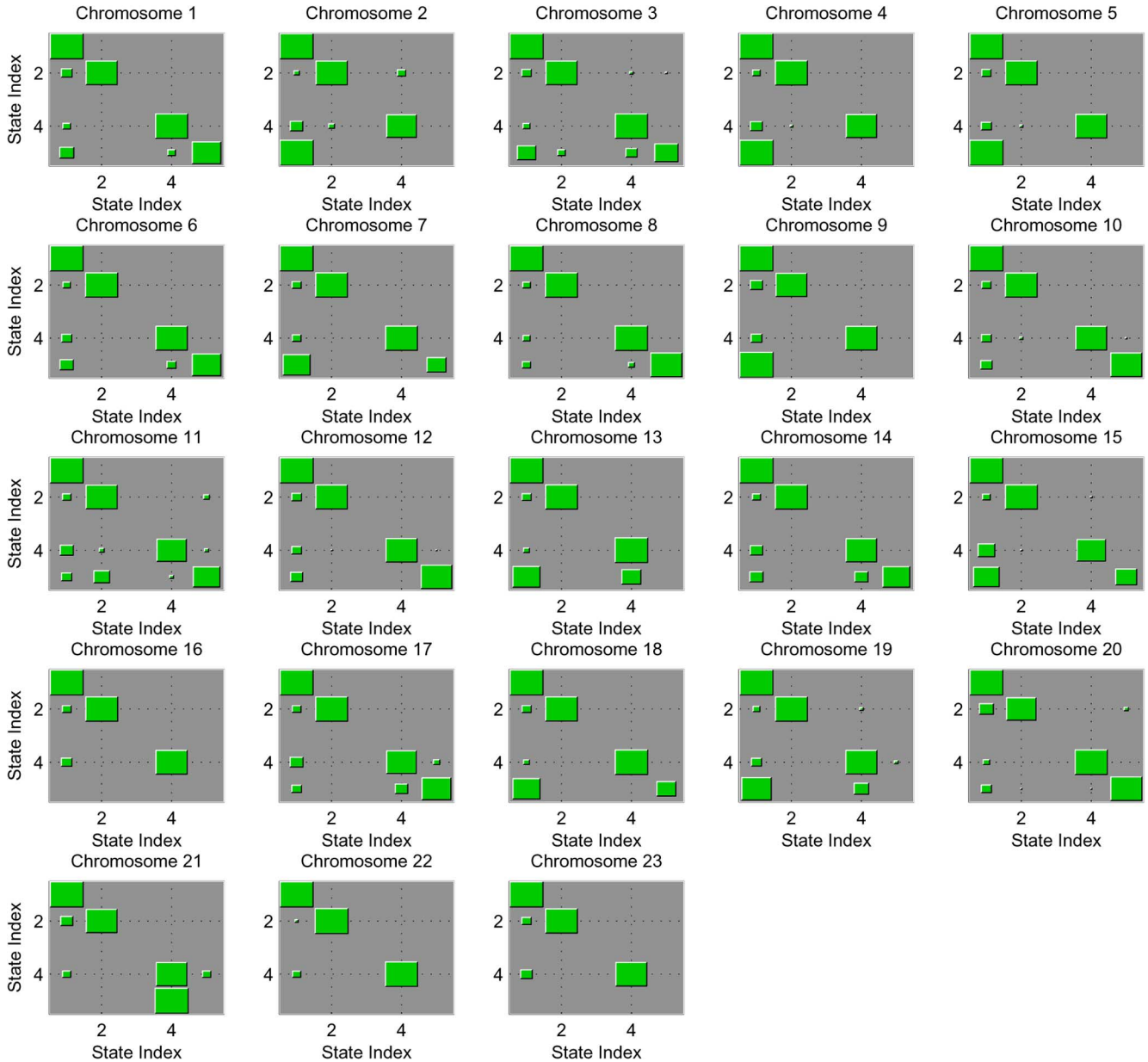


Fig. 5. Posterior expectation of state-transition matrices of each chromosome, inferred via the sticky DD-HMM. The plots are meant to visualize matrices, where the green box reflects the relative probability strength (where there is no green square, the probability is zero). In the model a state truncation level of $J = 20$ was employed. Nevertheless, since four meaningful states (state 1, state 2, state 3 and state 5) are inferred, here we only show the state-transition probabilities across the first 5 states, and those of other states, which are zeros, are omitted for clarity.

not considered in [15]. Since the four states are fixed, MCMC results via the Bayesian HMM [15] yield state labels based on collected samples; for the other two models considered (DD-HMM and sticky DD-HMM), for which VB inference is employed, a “hard” (most likely) decision is employed to label the hidden states, as discussed in Section V. Fig. 6 presents the state labels for example array-CGH profiles, as inferred by the three models. The horizontal axis in these plots denotes the index of the DNA fragments or “clones.”

From Fig. 6, we make the following *subjective* observations, with more quantitative results presented below. The model developed in [15], corresponding to (a) in Fig. 6, *a priori* imposes four states. It appears that this model is not particularly discriminating in the underlying states across the chromosome. For ex-

ample, consider sample s0177 of chromosome 8 (the right-most sample in the bottom row). Although the CGH value appears to noticeably variation with CGH position index (across the horizontal axis), almost all CGH values associated with this chromosome and sample are assigned to the same state. Similar phenomenon is exhibited on other samples.

By contrast, the DD-HMM model, corresponding to Fig. 6(b), is far more discriminating in the underlying states (more state diversity, in general, as a function of position index). However, because this model is not sticky, there is quick changing of states even where the CGH values are not changing significantly in strength (e.g., larger position indexes for sample b0338 of chromosome 8, the two from left sample in the bottom row). This phenomenon appears to manifest redundant (superfluous) states

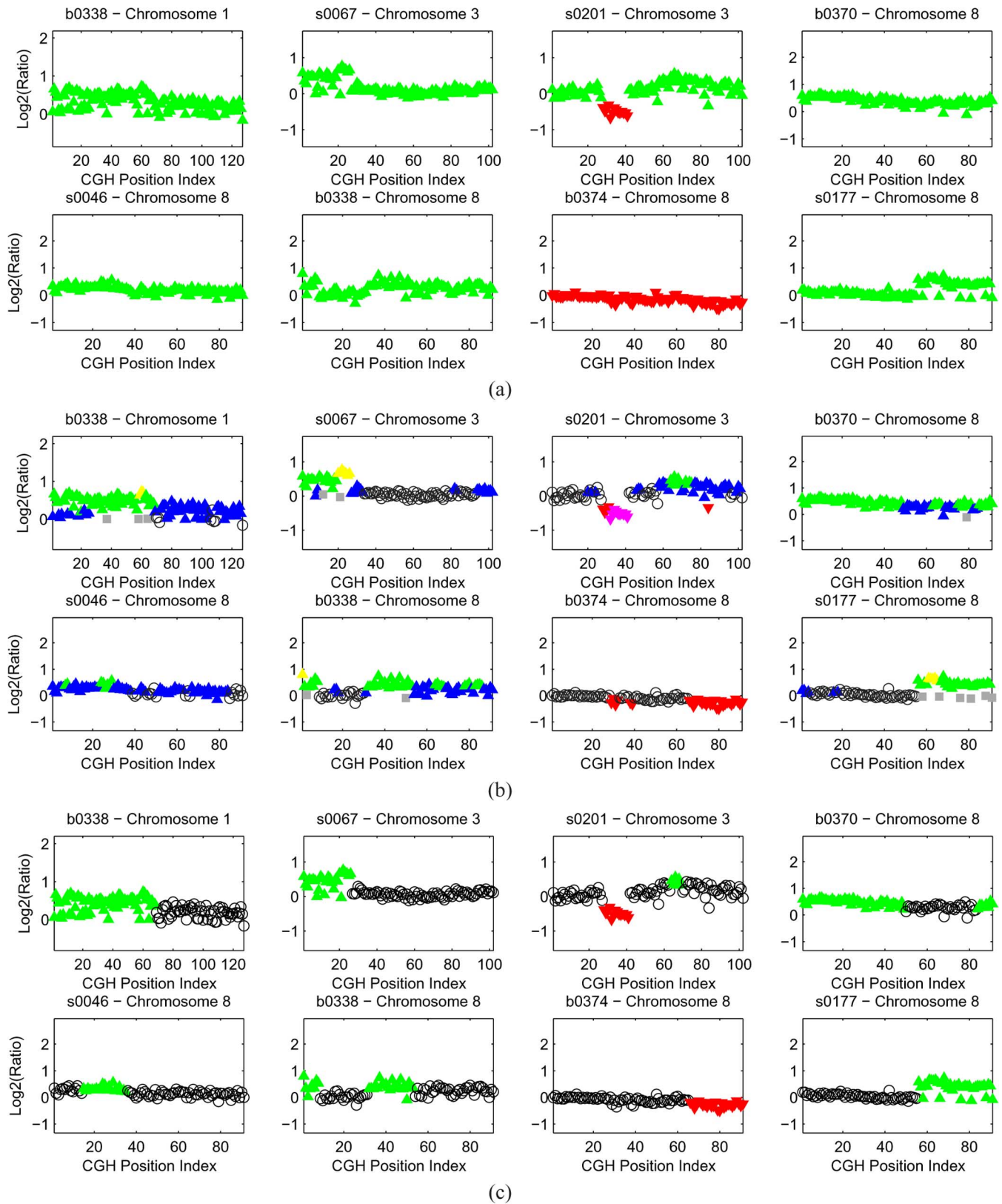


Fig. 6. State labels for example array-CGH profiles. (a) Bayesian HMM [15], (b) DD-HMM, (c) Sticky DD-HMM. The colored symbols denote different states: one normal state inferred via Bayesian HMM and sticky DD-HMM, represented by black $\circ$, and two inferred via DD-HMM, represented by black $\circ$ and gray $\blacksquare$; one copy loss state inferred via Bayesian HMM and sticky DD-HMM, represented by red $\blacktriangledown$, and two inferred via DD-HMM, represented by red and magenta $\blacktriangledown$; one small copy gain state inferred via Bayesian HMM and sticky DD-HMM, represented by green $\blacktriangle$, and three inferred via DD-HMM, represented by green, blue and yellow $\blacktriangle$.

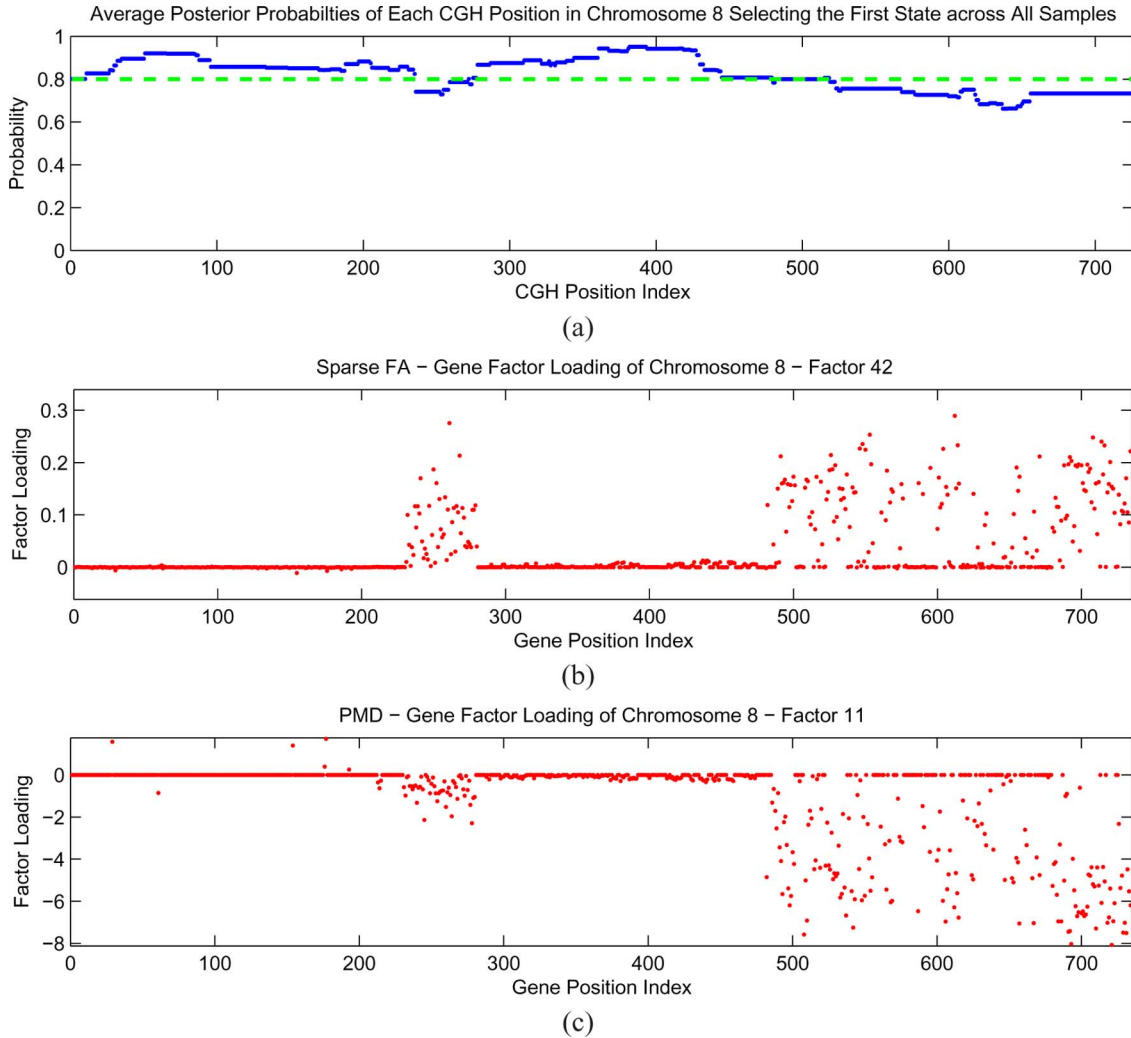


Fig. 7. (a) Posterior probabilities of each CGH position in chromosome 8 selecting the normal state, which shows the average probabilities across all sample; (b) Gene factor loadings in chromosome 8 of factor 42 via FA [6] with a total of $K = 55$ factors; (c) Gene factor loadings in chromosome 8 of factor 11 via PMD [35] with a total of $K = 55$ factors. The CGH and gene measurements are aligned according to the index of the DNA fragments or “clones.”

to mitigate the non-stickiness, thereby undermining model interpretability. This has motivated the proposed sticky HMM, and the large scale of this problem has motivated a formulation that admits a VB solution.

The proposed sticky HMM results are shown in Fig. 6(c). Note that this model yields a more diverse state usage than the model considered in Fig. 6(a), but the inferred states appear to manifest the desired stickiness. For example, sample s0201 of chromosome 3 (three from left, top row) manifests clear regions defined by three different states, contiguously partitioned as a function of position index. The states appear to capture well the relative CGH intensity. Returning to sample b0338 of chromosome 8, note that all the CGH values beyond index 60 along the horizontal are assigned to a single state [unlike the results in Fig. 6(b)].

There is no explicit “ground truth” available for the CGH data, and therefore we use an associated but independent data set based on gene-expression values on the same samples. Further, to analyze these gene-expression data we employ two established methods: a Bayesian sparse factor analysis (FA) model [6] and the related (but non-Bayesian) PMD method developed in [35].

The gene-expression data $\mathbf{Y} \in \mathbb{R}^{N \times L}$ is normalized to zero mean in each row, where each row represents a separate gene, and the columns correspond to different samples. The FA model seeks to factorize the data matrix into the form $\mathbf{Y} = \mathbf{AS} + \mathbf{E}$, where $\mathbf{A} \in \mathbb{R}^{N \times K}$ is the factor loading matrix, $\mathbf{S} \in \mathbb{R}^{K \times L}$ is the factor matrix, each row being a factor. Here N is the number of genes and K is the number of factors. Typically one simply sets $K[6]$, with $K \ll N$; $\mathbf{E} \in \mathbb{R}^{N \times L}$ is the error/noise matrix, addressing those aspects of $\mathbf{Y}$ not captured in the factors. For the data considered $N = 22215$ and $L = 89$.

For gene-expression data analysis, one usually imposes sparseness priors on the factor loading matrix, shrinking most of the elements to be near zero. A “spike-slab” sparseness construction is used in [6]. As discussed in [6], since factor loading $\mathbf{A}$ is “sparse,” which means many of the elements of $\mathbf{A}$ are close to zero, each column ideally will represent a particular biological “pathway,” composed of a relatively small number of relevant genes related to a given latent factor, which correspond to those having factor loadings not close to zero. In contrast, PMD employs a non-Bayesian method to achieve the same goal. In the PMD method, the matrix $\mathbf{Y}$ is approximated as $\hat{\mathbf{Y}} = \sum_{k=1}^K \xi_k \mathbf{U}_k \mathbf{\Lambda}_k$ by minimizing $\|\mathbf{Y} - \hat{\mathbf{Y}}\|_F^2$ subject

to sparseness penalties on $\mathbf{U}_k$ and $\mathbf{\Lambda}_k$. When the PMD is applied using an ℓ_1 penalty on $\mathbf{U}_k$ but not on $\mathbf{\Lambda}_k$, a method for sparse principle components results. In this way, $\xi_k \mathbf{U}_k$ corresponds to factor loading of factor k , and $\mathbf{\Lambda}_k$ corresponds to factor score.

In the two factor models, the (sparse) factor-loading matrix $\mathbf{A}$ yields the (typically relatively small) subset of genes associated with a given “pathway,” and these genes may be linked to regions on the 23 chromosomes. The objective is to examine whether the genes responsible for the inferred pathways reside at portions of the chromosomes at which “interesting” activity (raised or lowered level of copying) is revealed via the aforementioned CGH analysis.

Fig. 7 presents results in which the gene-expression analysis was performed via FA [6] and PMD [35]. The Fig. 7(a) display posterior probabilities of each CGH position in chromosome 8 selecting the normal state (State 1). Specifically, the average probabilities across all samples are $(1/L) \sum_{l=1}^L \langle s_{t,1}^{(m,l)} \rangle$, where $\langle s_{t,1}^{(m,l)} \rangle$ represents the posterior expectation of state indicator $s_t^{(m,l)}$ with outcome 1, and parameter $L = 89$ denotes the total number of samples. The lower the probability of a CGH being in State 1, presumably the more important the CGH measurements are, since the CGH fragments with copy number losses or gains might be relevant to the breast cancer associated with these data. In Fig. 7(b) and (c), we plot the factor loadings in chromosome 8 of factor 42 via FA and those of factor 11 via PMD, which provides the principal description of the gene-expression data in this chromosome. Here we set a total of $K = 55$ factors for both models. From Fig. 7, we observe that the positions of informative (non-State 1) CGH data and inferred important genes are consistent. Specifically, those regions of the chromosome that have a low probability of being in State 1 (lower than 0.8) are also regions (generally) for which the associated genes contribute to the factor 42 via FA and factor 11 via PMD. Very similar results were found for other chromosomes, and are therefore omitted for brevity.

These results appear to suggest that CGH copy numbers that are inconsistent with typical behavior (*not* in State 1) are indicative of a portion of the chromosome that are linked to the illness under study, here breast cancer. Note that the ability to explicitly localize State 1 is therefore important, with this a challenge for the *non-sticky* HMM, since redundant states may be manifested, undermining interpretability of the results [see Fig. 6(b)].

Considering the diagnosis of algorithm convergence from the log model evidence [lower bound shown in (16)], if we use the lower bound variation rate between two iterations lower than 10^{-5} as the convergence threshold, the VB algorithm was found to converge after typically 50 iterations, as shown in Fig. 8; if we use lower threshold, similar results were found for larger iterations.

All the above sticky DD-HMM results are based on truncation level $J = 20$ and hyper-parameter $\alpha/J = 10^{-3}$. Fig. 9(a) and (b) present the variation of the number of inferred states with different truncation levels and α/J separately. The results demonstrate the number of states is not sensitive to

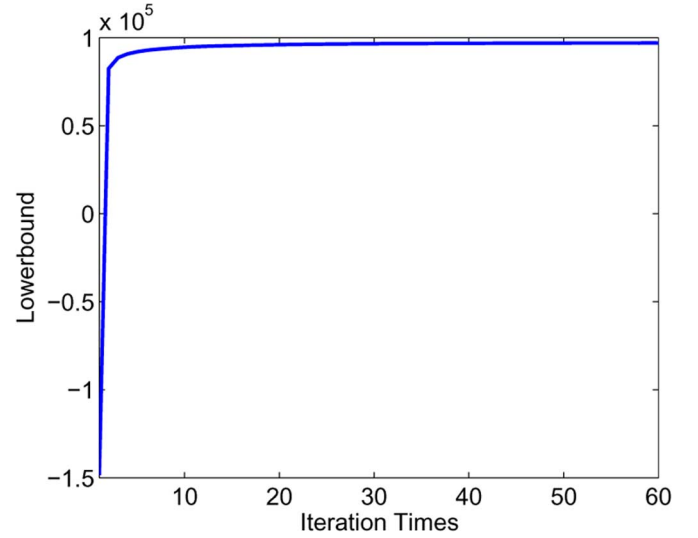


Fig. 8. VB iteration lowerbound for MTL sticky DD-HMM on the real CGH data.

the truncation level, and for each J the $\mathbf{w}$ and $\mathbf{w}_0$ are sparse. However, from Fig. 9(b) there is a large range of α/J for which the model infers the same number of underlying states.

All experiments have been performed in non-optimized software written in Matlab, on a Pentium PC with 1.73 GHz CPU and 4 G RAM. One VB run of the MTL sticky DD-HMM with truncation level $J = 20$, for 60 VB iterations, required about 1.5 hours for the whole CGH dataset (processing all chromosomes simultaneously). It only required about 20 seconds for each Bayesian HMM [15] MCMC run when considering a single CGH profile of one chromosome; however, all runs for all 89 CGH profiles of all 23 chromosomes required more than 11 hours. In addition to the computational challenges associated with Bayesian HMM [15], each CGH sample of each chromosomes is analyzed in isolation, and therefore multi-task learning is not implemented (the state statistics are not explicitly shared across chromosomes). Further, the results in Fig. 6 suggest inferior underlying state inference (which may be attributable to the lack of multi-task analysis).

VII. CONCLUSION

We have developed a hidden Markov model (HMM) with state persistence, termed the sticky HMM with Dirichlet distribution prior (sticky DD-HMM). The new model is motivated by [10] and [11], and the proposed construction allows convenient VB inference, of interest for the large-scale motivating CGH problem. For array CGH data analysis, we further extended traditional single-task HMM to multi-task learning (MTL), where here the multiple tasks are linked to specific chromosomes. The proposed multi-task model allows simple VB inference, yielding fast computation times and efficient detection of copy losses and gains. The algorithm has been demonstrated on synthetic data and real CGH data. The CGH results are partially validated by a corresponding gene analysis using factor models [6], [35]. The sticky DD-HMM extends the Bayesian HMM for application to CGH [15], in that stickiness is explicitly

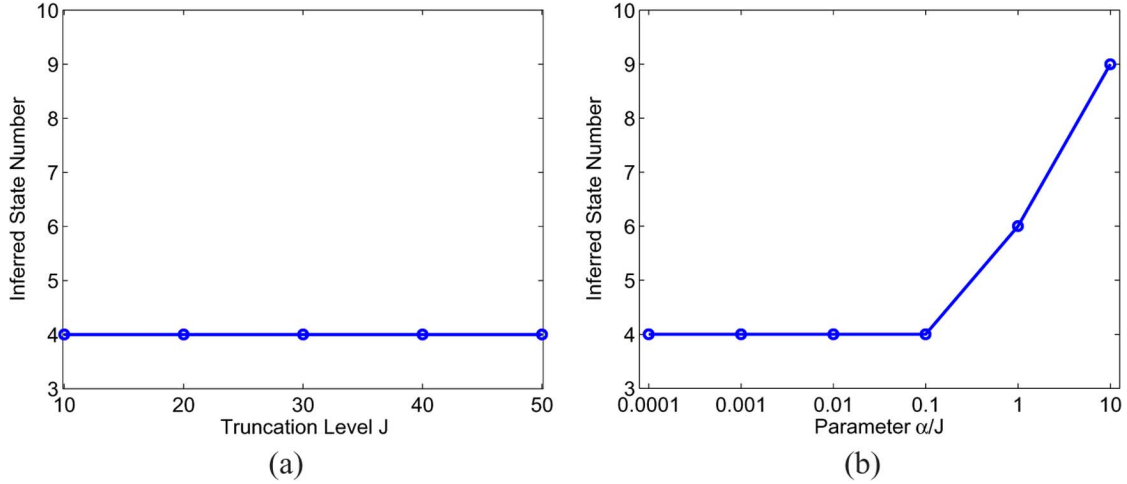


Fig. 9. (a) Variation of the number of inferred states with different truncation levels, for $\alpha/J = 10^{-3}$. (b) Variation of the number of inferred states with different hyper-parameter α/J , for $J = 20$.

imposed, and the number of underlying states is inferred from the data.

APPENDIX

DERIVATION OF UPDATE EQUATIONS IN VB APPROACH

For the model introduced in Section III and the corresponding mean-field variational distribution described in (17), $q(\theta_j)$ is defined by the specific application.

$$\log q(\theta_j) \propto \log H(\theta_j | \tilde{\theta}_j) + \sum_{t=1}^T \langle s_{t,j} \rangle \log f(x_t | \theta_j); \quad j = 1, \dots, J \quad (21)$$

where $\langle s_{t,j} \rangle$ denotes the expected number of state indicator s_t with outcome j . If the prior $H(\cdot)$ is conjugate to the likelihood $f(\cdot)$ then we can easily get the update equation for $q(\theta_j)$.

For updating $q(\mathbf{w}_0)$ and $q(\mathbf{w}_j)$ given the prior $p(\mathbf{w}_0) = \text{Dir}(\alpha/J, \dots, \alpha/J)$ and $p(\mathbf{w}_j) = \text{Dir}(\alpha/J, \dots, \alpha/J)$, assume $q(\mathbf{w}_0) = \text{Dir}(\tilde{\alpha}_0)$ and $q(\mathbf{w}_j) = \text{Dir}(\tilde{\alpha}_j)$. Then

$$\begin{aligned} \tilde{\alpha}_{0,i} &= \frac{\alpha}{J} + \langle s_{1,i} \rangle; \quad i = 1, \dots, J \\ \tilde{\alpha}_{j,i} &= \frac{\alpha}{J} + \sum_{t=1}^{T-1} \langle s_{t,j} \rangle \langle s_{t+1,i} \rangle \langle z_{t,1} \rangle; \\ i &= 1, \dots, J; \quad j = 1, \dots, J \end{aligned} \quad (22)$$

where $\langle z_{t,b} \rangle$ denotes the expected number of binary switch indicator z_t with outcome b , with $b = \{0, 1\}$.

Assume $q(\beta_j) = \text{Beta}(\tilde{c}_j, \tilde{d}_j)$, the update equations for $q(\beta_j)$ are as follows

$$\begin{aligned} \tilde{c}_j &= c_0 + \sum_{t=2}^T \langle s_{t-1,j} \rangle \langle z_{t,1} \rangle, \\ \tilde{d}_j &= d_0 + \sum_{t=2}^T \langle s_{t-1,j} \rangle \langle z_{t,0} \rangle; \quad j = 1, \dots, J. \end{aligned} \quad (23)$$

Given the approximate distribution of the other variables,

$$\begin{aligned} q(\mathbf{s}) \propto \exp \left[\langle \log w_{0,s_1} \rangle + \left[\sum_{t=1}^{T-1} \langle z_{t+1,0} \rangle \langle \log u_{s_t, s_{t+1}} \rangle \right] \right. \\ \left. + \left[\sum_{t=1}^{T-1} \langle z_{t+1,1} \rangle \langle \log w_{s_t, s_{t+1}} \rangle \right] \right. \\ \left. + \left[\sum_{t=1}^T \langle \log f(x_t | \theta_{s_t}) \rangle \right] \right] \end{aligned} \quad (24)$$

where $\langle \cdot \rangle$ represents the expectation of the associated variable's distribution. One may readily derive that

$$\begin{aligned} \langle \log w_{i',j} \rangle &= \psi(\tilde{\alpha}_{i',j}) - \psi \left(\sum_{h=1}^J \tilde{\alpha}_{i',h} \right); \\ i' &= 0, \dots, J; \quad j = 1, \dots, J \end{aligned} \quad (25)$$

in which $\psi(\cdot)$ is the Digamma function. In (24), $\langle \log f(x_t | \theta_j) \rangle$ is the data likelihood, with expectation performed with respect to the distribution of atom θ_j (which may be derived readily for the specific application).

Similarly,

$$\begin{aligned} q(z_t = 0) &\propto \exp \left[\langle \log(1 - \beta_{s_t}) \rangle + \langle \log u_{s_{t-1}, s_t} \rangle \right. \\ &\quad \left. + \langle \log f(x_t | \theta_{s_t}) \rangle \right]; \quad t = 2, \dots, T \\ q(z_t = 1) &\propto \exp \left[\langle \log \beta_{s_t} \rangle + \langle \log w_{s_{t-1}, s_t} \rangle \right. \\ &\quad \left. + \langle \log f(x_t | \theta_{s_t}) \rangle \right]; \quad t = 2, \dots, T \end{aligned} \quad (26)$$

where $\langle \log(1 - \beta_j) \rangle = \psi(\tilde{d}_j) - \psi(\tilde{c}_j + \tilde{d}_j)$ and $\langle \log \beta_j \rangle = \psi(\tilde{c}_j) - \psi(\tilde{c}_j + \tilde{d}_j)$ with $j = 1, \dots, J$.

Note that when computing $\log u_{j,i}$ in (24) and (26), to avoid $\log 0$, an approximation method is needed. In practice, we substitute a very small positive number ϵ for each zero and $1 - (J - 1)\epsilon$ for the one in each $\mathbf{u}_j$.

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